NETWORKING ASSIGNMENT

VLAN CONFIGURATION AND VERIFICATION

Topics: VLAN creation, port assignment, connectivity testing, and automation

NETWORK

Name: _____

Class / Roll No.: _____

Date: _____

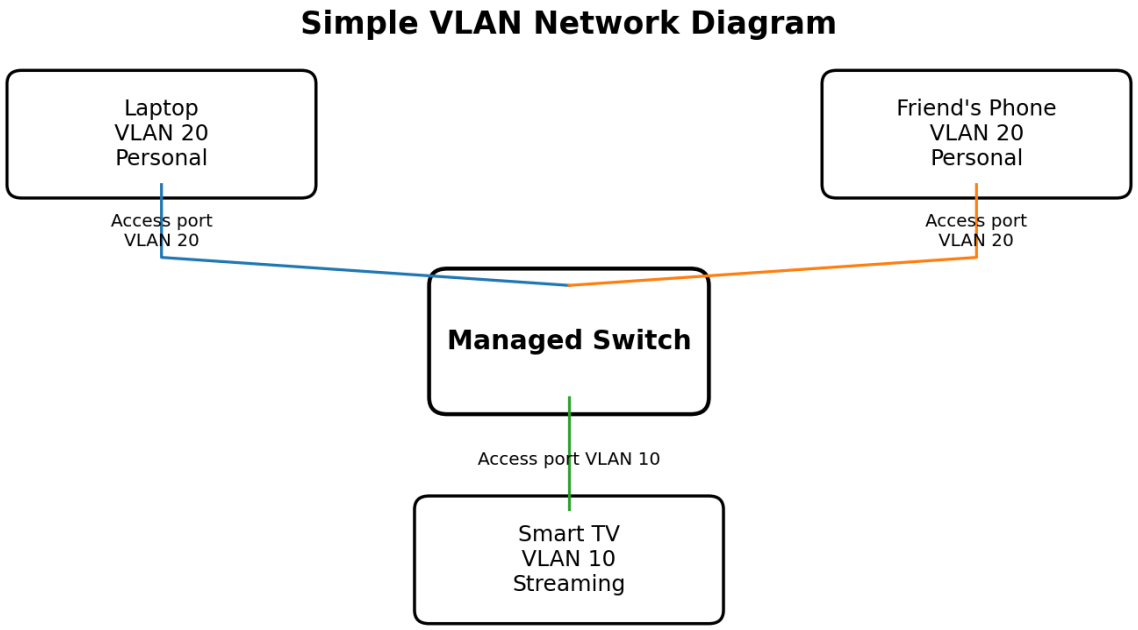
Task 1: VLAN Network Diagram

QUESTION

Draw a simple network diagram showing three devices (a laptop, a friend's phone, and a smart TV) connected to a switch, and label each device's VLAN (VLAN 10 for streaming and VLAN 20 for personal devices).

ANSWER

The diagram below shows a managed switch connecting three devices. The smart TV is placed in VLAN 10 for streaming, while the laptop and friend's phone are placed in VLAN 20 for personal devices.



VLAN 10 and VLAN 20 are separate Layer 2 broadcast domains. Devices in different VLANs normally cannot communicate directly unless a router or Layer 3 switch is configured to route between them.

Task 2: Create VLANs and Verify Connectivity

QUESTION

In Cisco Packet Tracer or NetSim, create VLAN 10 and VLAN 20, assign three PCs to VLAN 10 and two PCs to VLAN 20, and verify that PCs in different VLANs cannot ping each other.

ANSWER

Example topology: connect five PCs to switch ports Fa0/1–Fa0/5. Assign PCs 1–3 to VLAN 10 and PCs 4–5 to VLAN 20. Use IP addresses in the same subnet for a basic Layer 2 isolation test.

```
enable
configure terminal
vlan 10
 name STREAMING
exit
vlan 20
 name PERSONAL
exit
interface range fastEthernet 0/1-3
 switchport mode access
 switchport access vlan 10
exit
interface range fastEthernet 0/4-5
 switchport mode access
 switchport access vlan 20
exit
end
show vlan brief
```

Example PC addressing: PC1 192.168.10.11/24, PC2 192.168.10.12/24, PC3 192.168.10.13/24; PC4 192.168.20.11/24, PC5 192.168.20.12/24. Keep each PC's default gateway blank for this Layer 2 test.

Verification: run `show vlan brief` and confirm Fa0/1–Fa0/3 are in VLAN 10 and Fa0/4–Fa0/5 are in VLAN 20. Ping between two PCs in the same VLAN should succeed when cabling and IP settings are correct. Ping from a VLAN 10 PC to a VLAN 20 PC should fail because inter-VLAN routing is not configured. Record the actual simulator results in your submission.

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Task 3: Move a PC from VLAN 10 to VLAN 20

QUESTION

In a network simulator, move a PC from VLAN 10 to VLAN 20. Show the before-and-after configuration commands and test connectivity to confirm the change.

ANSWER

Assume PC3 is connected to switch port Fa0/3. Initially, Fa0/3 belongs to VLAN 10. To move PC3 to VLAN 20, change the access VLAN on that port.

Before configuration (PC3 in VLAN 10):

```
enable
configure terminal
interface fastEthernet 0/3
 switchport mode access
 switchport access vlan 10
end
show vlan brief
```

After configuration (PC3 moved to VLAN 20):

```
enable
configure terminal
interface fastEthernet 0/3
 switchport mode access
 switchport access vlan 20
end
show vlan brief
```

Connectivity test: give PC3 an address in the VLAN 20 subnet, for example 192.168.20.13/24, if the lab uses separate IP subnets for each VLAN. Ping a reachable PC in VLAN 20, such as 192.168.20.11; it should succeed if settings are correct. A ping to a VLAN 10 PC should fail without inter-VLAN routing. If the PC keeps an address from the old subnet, update its IP configuration before testing.

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Task 4: Automate Assigning 10 Ports to VLAN 30

QUESTION

Write a short script or command list to automate assigning 10 switch ports to VLAN 30, using Packet Tracer, NetSim, or GNS3 syntax. Use a loop or range command if supported to avoid repeating the command 10 times.

ANSWER

Cisco IOS-style CLI supports `interface range`, which selects multiple ports at once. The example below assigns FastEthernet ports Fa0/1 through Fa0/10 to VLAN 30.

```
enable
configure terminal
vlan 30
 name VLAN30
exit
interface range fastEthernet 0/1-10
 switchport mode access
 switchport access vlan 30
exit
end
show vlan brief
```

Expected result: `show vlan brief` should list Fa0/1–Fa0/10 under VLAN 30. This range command avoids entering the same configuration separately on each port. If the simulator or switch uses GigabitEthernet ports, substitute the correct interface type and numbering.

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